Name:_____ Period:_____

Why Do Astronauts Float?

Learning Intention: Today we will be learning why astronauts float in space so we can explain how gravity keeps objects in orbit.

Success Criteria:
- I can explain why objects fall at the same rate when there is no air.
- I can describe the two things an object needs to stay in orbit.
- I can use evidence to explain why astronauts float.

Part 1: Rise

This is **Rise**, a small plush toy that flew on **Artemis II** in April 2026. Plush toys ride along on spacecraft as "zero-gravity indicators." Once the spacecraft reached space, Rise started **floating** around the cabin.

1. Why do you think Rise floats? Circle one.
A. There is no gravity in space.
B. The spacecraft is too far from Earth for gravity to work.
C. Rise and the spacecraft are falling.
D. Something else: _____

2. Why did you pick that answer?

_____

_____

Part 2: Hammer and Feather

3. Predict: An astronaut on the Moon drops a hammer and a feather at the same time. Which lands first?

_____

4. What happened in the video?

_____

5. Watch the paper and book drop in class.

a. What happened when the paper fell by itself? _____

b. What happened when the paper fell on top of the book? _____

6. The hammer and feather landed together on the Moon, but the paper fell slowly by itself here. What is different about the Moon? (Think about what surrounds the falling objects.)

_____

_____

Part 3: The Falling Cup

7. Predict: The cup is leaking water. What will happen to the leak when the cup is dropped?

_____

8. What actually happened?

_____

9. Why do you think the water stopped coming out? (Think about what the cup and the water were both doing.)

_____

_____

Part 4: PhET Gravity and Orbits

Open **PhET Gravity and Orbits** and choose the **Model** screen with the Sun and Earth. Run each test below and sketch the path Earth takes. To change Earth's speed, pause the sim first, then drag the end of Earth's velocity arrow.

10. Run each test.

Test	Sketch the path	What happened?
Normal (gravity on, normal speed)		
Gravity off		
Gravity on, no speed (drag the velocity arrow to zero)		

11. Challenge: Change Earth's starting speed. Find a speed that makes Earth crash into the Sun, one that makes it orbit, and one that makes it escape.

Outcome	Was the speed slow, medium, or fast?	Sketch the path
Crash		
Orbit		
Escape		

12. To stay in orbit, an object needs two things.

a. _____

b. _____

Part 5: Explain It

13. Why does Rise float? Use at least two pieces of evidence from today.

Claim: _____

Evidence 1: _____

Evidence 2: _____

14. Exit question: Did your answer to Question 1 change? What changed your mind, or what made you keep it?

_____

_____